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A void-based framework for predicting hydrogen embrittlement

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ABSTRACT

Hydrogen embrittlement alters defect activity, interfacial cohesion, and fracture processes in structural metals, yet predictive links between hydrogen–defect interactions and fracture toughness remain limited. The role of crack-tip constraint under hydrogen exposure is particularly unclear due to the coupled effects of mechanical fields, hydrogen transport, and competing embrittlement mechanisms. Here we develop H-CGM+, a mechanism-informed, void-based predictive framework that integrates hydrogen-enhanced localized plasticity (HELP), hydrogen-enhanced strain-induced vacancy formation (HESIV), and hydrogen-enhanced decohesion (HEDE), together with strain-dependent hydrogen trapping. A central advance is the introduction of a hydrogen-enhanced void nucleation (HEVN) mechanism, which links HELP and HESIV and provides a physically grounded description of hydrogen-suppressed ductile fracture. The framework captures the competition between ductile and brittle fracture modes under hydrogen within a unified formulation. Systematic analyses using a generic BCC material dataset show that increased crack-tip constraint continues to reduce fracture toughness, consistent with behavior in air, within the current mechanistic description. However, the model does not reproduce the emerging, yet not well-established, trend in which hydrogen may reverse this relationship. These results demonstrate both the predictive capability and current limitations of H-CGM+ and provide insights relevant to the design and integrity assessment of hydrogen infrastructure and structural components, highlighting key knowledge gaps and the need for targeted experimental and theoretical studies to advance predictive modeling of hydrogen-assisted fracture.

1. Introduction

The widespread adoption of hydrogen as a clean energy vector has renewed the urgency to understand and mitigate hydrogen embrittlement (HE) in metallic materials, particularly in high-strength pipeline steels [1,2]. HE remains one of the most complex challenges in materials science, resulting from the intricate interplay between fracture driving forces and the material's resistance, initially governed by its microstructure and subsequently weakened by the presence of hydrogen [1].

HE encompasses a broad spectrum of fracture modes, ranging from hydrogen-suppressed ductile fracture characterized by microvoid nucleation, growth, and coalescence (commonly abbreviated as MVC for brevity) at low hydrogen concentrations, to hydrogen-assisted quasi-cleavage (QC) fracture [3] at higher hydrogen concentrations. Notably, despite the common association of the term “embrittlement” with brittle failure, HE does involve ductile fracture, which can appear somewhat counterintuitive. MVC is a higher-level failure mechanism that can be linked to two fundamental HE mechanisms: hydrogen-enhanced localized plasticity (HELP) [4] and hydrogen-enhanced strain-induced vacancy (HESIV) [5]. In some materials, further increases in hydrogen concentration may lead to intergranular (IG) fracture. Both QC and IG fractures are called hydrogen-assisted brittle fracture in this study and typically associated with the Hydrogen-Enhanced Decohesion (HEDE) mechanism. However, the correspondence between fracture morphology and mechanism is not one-to-one. In this paper, we refer to brittle fracture to denote the stress-controlled decohesion-type failure process, but the damage or material degradation leading to it is frequently an interplay with HELP/HESIV [6]. For example, plasticity-driven, decohesion-related transgranular fracture has been reported in the literature [7,8], supporting the mechanisms synergy

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independently of the fracture morphology.

Fig. 1(a) schematically illustrates this spectrum of HE behavior in a tensile bar as a function of hydrogen concentration. It should be noted that not all materials exhibit the full range of fracture behavior; for example, pipeline steels typically show hydrogen-suppressed ductile fracture and hydrogen-assisted QC mode. The fracture surfaces of a typical pipeline steel failed in zones 1, 2 and 3 are shown in Fig. 1(b). A thorough discussion of the independence and synergistic interactions of the fundamental mechanisms can be found in [12].

Accurate prediction of HE as a function of mechanical loading, hydrogen environment, and microstructural parameters is critical for ensuring the structural integrity of hydrogen transport and storage systems [13]. In this context, a "predictive model" is referred to as a mechanistic model capable of forecasting the critical conditions leading to crack initiation, based on these coupled effects. Such a model should accurately represent the underlying failure mechanisms and ultimately enable the transfer of model parameters from controlled laboratory experiments to the integrity assessment of components under service conditions. Despite the significant amount of microscopic insights into HE mechanisms achieved during decades of research, it has been noted that "possibly the most crucial shortcoming of the HE field is a dearth of predictive models" [14]. This shortcoming largely stems from the fact that the fundamental mechanisms are inherently descriptive in nature and not straightforward in relation to fracture. Their qualitative frameworks, while crucial for mechanistic understanding, present substantial challenges for predictive model development.

Some recent progress has nevertheless been made in the predictive modeling of hydrogen-assisted fracture. At the level of fracture mode and constitutive failure criterion, these models can be broadly grouped into two principal classes: brittle fracture-based and ductile fracture-based approaches. Depending on the adopted interpretation framework for HE, different mechanisms may then be invoked separately or combined to rationalize how hydrogen activates or modifies these two failure modes [15]. For instance, decohesion-dominated, MVC-dominated, and HELP+HEDE interpretations can be viewed as different mechanistic routes through which hydrogen affects brittle fracture, ductile fracture, or their synergistic action [6,7]. A fully unified model that incorporates both brittle and ductile failure responses within the same micromechanically based constitutive relation, however, remains rare. This point is elaborated below, leading to the motivation for the

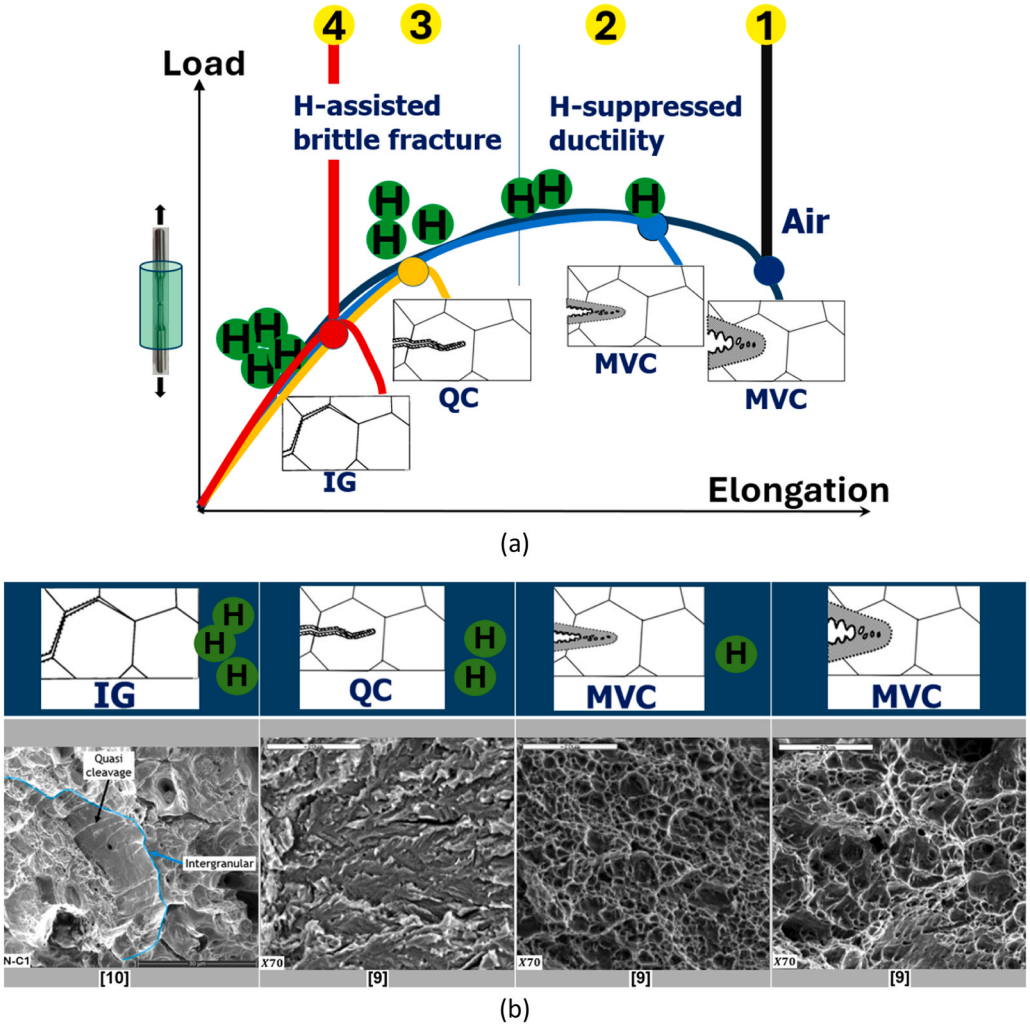


Fig. 1. (a) Schematic representation of HE as a function of hydrogen concentration, illustrating progressive ductility loss in a tensile specimen. In air (Zone 1), fracture is fully ductile, driven by MVC, representing the upper bound of ductility. Upon hydrogen charging, ductile fracture by MVC remains dominant in Zone 2. With increasing hydrogen concentration, fracture mode shifts from MVC to quasi-cleavage (QC, Zone 3), and eventually to intergranular fracture (IG, Zone 4), depending on microstructure and loading. Zones 1–3 exhibit transgranular fracture, while Zone 4 marks the lower bound of ductility under severe HE. (b) SEM images of X70 pipeline steel and N-C1 pipeline steel fracture surfaces from [9] and [10]. They show respectively the evolution of fracture surface from MVC in air to MVC in hydrogen with smaller and shallower dimples, followed by QC and IG under hydrogen. The graphical representations of the MVC, QC, and IG were redrawn, inspired by Beachem [11].

present model development.

Brittle fracture-based models include the cohesive zone model (CZM) [16] and the phase-field method (PFM) [17]. These approaches are typically applicable to Zones 3 and 4 in Fig. 1. Both rely on a fracture energy parameter which decreases with increasing hydrogen concentration. A hydrogen degradation law, derived from density functional theory (DFT) calculations [18], has been widely adopted in both CZM and PFM frameworks. Compared to CZM, PFM offers distinct advantages: it does not require a predefined crack path and can conveniently simulate multiple crack branches. Some recent works have extended PFM to include plasticity with various formulations [19]. These models either do not couple damage and plasticity, considering plasticity in the stress calculation while damage remains governed by elastic deformation [20,21], or they do couple them, incorporating hardening and dissipation terms into the damage energy balance [22].

Another type of predictive model is based on ductile fracture mechanisms. These models are particularly effective for simulating fracture behavior in Zone 1, which is governed by MVC. Among them, micromechanical models such as the Gurson-Tvergaard-Needleman (GTN) model have been widely used to represent these processes [23–25]. An extension of this framework, the Complete Gurson Model (CGM) proposed by the corresponding author, incorporates a localized deformation mode into the original GTN model which was derived from a homogeneous deformation mode, offering improved predictive capabilities for ductile failure [26–30]. A distinct feature of CGM is that failure arises naturally from the interaction between the two deformation modes, making void nucleation the root cause of material failure. This also means that CGM provides a direct link between the parameters governing void nucleation and macroscopic ductile failure behavior.

To extend ductile fracture modeling from Zone 1 to Zone 2, where hydrogen begins to suppress ductility, the CGM framework has been further augmented to incorporate hydrogen effects, resulting in the so-called hydrogen-extended CGM (H-CGM) [31,32]. Additionally, it accounts for stress-assisted hydrogen diffusion and plasticity-driven trap generation [31,33,34]. However, in the proposed preliminary H-CGM model, an ad hoc treatment of the effect of hydrogen was made without a sound physical basis.

It can be observed that the brittle and ductile fracture-based approaches converge in the middle of Fig. 1(a). Given the robustness of the H-CGM framework, a natural question arises: Can a ductile fracture-based model like H-CGM be extended to capture the full spectrum HE, encompassing both hydrogen-assisted brittle and hydrogen-suppressed ductile fracture? To address this, a hydrogen-assisted brittle failure mode has been incorporated into the original H-CGM framework, resulting in the so-called H-CGM+. This H-CGM+ allows the competition of the hydrogen-suppressed ductile fracture and hydrogen-assisted brittle failure and enables the prediction of fracture behavior across the entire HE spectrum, from Zone 1 to Zone 4. In this study, a physically sound H-CGM+ is developed and employed for the first time to predict fracture toughness under hydrogen charging.

Over the past three decades, a significant advancement in fracture mechanics has been the understanding and quantification of crack tip constraint. For relatively thin-walled components such as pipelines, the inability to satisfy the small-scale yielding conditions required for standard fracture mechanics testing inevitably leads to a loss of crack tip constraint, making fracture toughness no longer a unique material property. Previous studies have demonstrated that crack tip constraint -primarily governed by the initial crack length (a) -to-width (W) ratio (a/W) and loading mode - strongly influences the apparent fracture toughness measured in air [35,36]. In general, specimens with higher crack tip constraint exhibit lower fracture toughness, and for the same crack geometry, bending loads produce lower toughness than tension. The single-edge notched tension (SENT) specimen has gained increasing acceptance for evaluating fracture resistance in pipeline applications. Finite element analyses have shown that by appropriately selecting the crack length-to-width ratio a/W, the crack tip constraint in SENT specimens can be adjusted to closely match, though remain slightly higher than, that of surface cracks in actual pipelines [37–40]. Due to this similarity in constraint levels, fracture toughness data obtained from SENT tests can be directly applied in structural integrity assessments of pipelines. Accordingly, industrial guidelines and standards have been established for fracture toughness testing using SENT specimens [41,42].

However, the role of crack tip constraint under hydrogen charging remains unclear. Compared to its well-documented effect on ductility, the influence of hydrogen on fracture toughness has received relatively limited attention. This is largely due to the high cost and experimental difficulty associated with determining fracture toughness and fracture resistance behavior in hydrogen environments. Fig. 2 illustrates the known effect of crack tip constraint on fracture toughness in air, along with three hypothesized scenarios under hydrogen charging. In Scenario H1, hydrogen reduces fracture toughness, but the trend with constraint remains like the one observed in air. In Scenario H2, hydrogen diminishes the influence of constraint, rendering fracture toughness relatively independent of constraint level. In Scenario H3, hydrogen causes a more pronounced reduction in fracture toughness at low constraint, such that fracture resistance becomes lower under low-constraint conditions than under high-constraint ones. Although systematic experimental studies are ongoing, some evidence suggests that Scenario H3 may be valid [43]. This has serious implications for the structural integrity of hydrogen pipelines as well as for the re-purposing of the existing gas pipelines for hydrogen transport. Current design and assessment methodologies use fracture toughness data obtained from SENT specimens, under the assumption that this provides a less conservative but still safe basis for safety. However, if hydrogen leads to lower fracture toughness under low-constraint conditions, this may have drastic consequences for fracture toughness testing in hydrogen environments. This suggests that SENT testing should be performed for materials designated for hydrogen pipelines, as such specimens better represent the stress states encountered in service. In contrast, using high-constraint specimens such as CT samples could lead to overestimated fracture toughness values and, consequently, unsafe structural assessments. If this is the case, there will be an urgent worldwide need to adopt SENT testing as a standard approach for evaluating and qualifying materials intended for hydrogen pipeline applications [44,45].

In this study, we develop and present the latest mechanism-informed

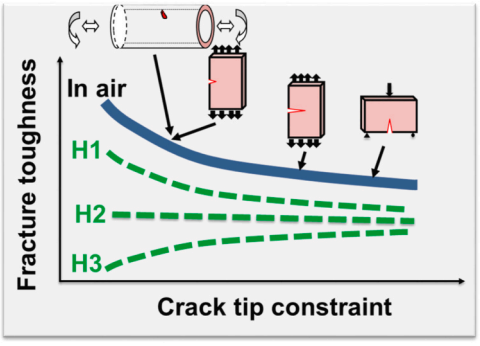


Fig. 2. Schematic illustration of the effect of crack tip constraint on fracture toughness in air and potential scenarios in a hydrogen environment. For SENT specimens, increasing the crack length-to-width ratio raises the crack tip constraint. A bending specimen usually has higher constraint than a corresponding tensile specimen. The constraint level in a SENT specimen can be tuned to be very close to that of surface cracks in a pipe. In air, fracture toughness decreases with increasing level of constraint. Under hydrogen, three possible trends are considered: (H1) hydrogen reduces fracture toughness, but the toughness–constraint relationship follows the same trend as in air; (H2) hydrogen neutralizes the effect of constraint, making fracture toughness relatively independent of it; and (H3) hydrogen causes a more significant reduction in fracture toughness at low constraint levels.

void-based predictive framework H-CGM+, which incorporates the current understanding of HE mechanisms and trapping behaviors, for the first time to investigate the influence of hydrogen on the relationship between fracture toughness and crack tip constraint. Using SENT specimens with systematically varied constraint levels under electrochemical hydrogen charging, we examine how geometric constraint, initial hydrogen concentration in lattice sites C_L^i, material-specific sensitivity parameters, and hydrogen trapping behaviors interact to influence fracture behavior. Specifically, the study aims to: (i) quantify the interaction between crack tip constraint and hydrogen on fracture resistance, (ii) predict the hydrogen-induced ductile-to-brittle transition under varying constraint conditions, and (iii) identify the plausible forms of the toughness–constraint relationship in the presence of hydrogen. The results provide novel mechanistic insights into the interplay between constraint, hydrogen, and fracture toughness, with important implications for the safe design of hydrogen transport infrastructure.

The structure of the paper is as follows. Section 2 presents the latest void-based predictive framework for HE, H-CGM+, with emphasis on its application to fracture toughness prediction, expressed in terms of crack tip opening displacement (CTOD), in SENT specimens. Section 3 outlines the material properties, microstructure-specific parameters, hydrogen charging conditions, trapping behaviors, and the numerical procedure, presents the results and discussion, followed by concluding remarks in Section 4.

2. A void-based predictive framework for HE: H-CGM+

In this study, we present the H-CGM+ framework, which is employed to predict fracture toughness across the full spectrum of HE, capturing the transition from ductile to brittle fracture. The framework integrates several key features, see Fig. 3: (1) a unique framework that allows the competition of hydrogen-suppressed ductile fracture and hydrogen-assisted brittle fracture; (2) a new mechanism-informed treatment of hydrogen effects, where hydrogen primarily influences void nucleation (hydrogen-enhanced void nucleation) through established HE mechanisms such as HELP and HESIV formation; (3) a chemical potential-based formulation for hydrogen diffusion, originally formulated by Díaz et al. [33], which avoids the need to compute hydrostatic stress gradients; and (4) an explicit time integration scheme with unlimited crack growth, selected for its effectiveness in capturing local instabilities and strong nonlinearities associated with damage evolution and crack propagation in the presence of hydrogen.

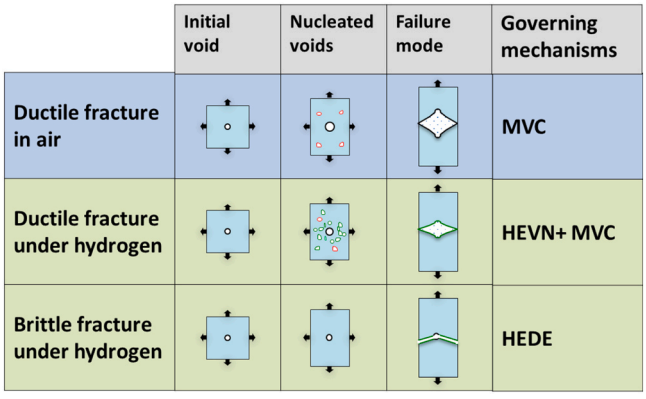


Fig. 3. Schematic illustration of the competing fracture modes considered in the present predictive framework, with ductile fracture in air included for comparison. The first column of figures depicts pre-existing voids (black), which are readily activated upon the onset of plastic deformation. The second column shows void nucleation during plastic deformation: red denotes voids formed in the absence of hydrogen, whereas green indicates hydrogen-enhanced void nucleation (HEVN). The third column summarizes the resulting fracture modes in air and hydrogen environments. Although pre-existing voids are also expected in brittle fracture, their role differs fundamentally: rather than contributing to microvoid coalescence (MVC), they act as stress concentrators that facilitate brittle failure.

2.1. Ductile fracture in air – the CGM model

The study of ductile fracture in metals, typically characterized by the sequence of MVC, is close to being a mature field. A major advancement came from Gurson, who derived a dilatational constitutive model by solving the plastic flow problem of a representative volume unit cell containing a pre-existing void. His model accounts for both the influence of voids on plastic flow and the effects of plastic deformation and stress triaxiality on void growth. It should be noted that Gurson model is fundamentally a void growth model based on the assumption of homogeneous plastic flow around voids. Tvergaard and Needleman [3] later extended Gurson's formulation by introducing a phenomenological treatment of void coalescence, resulting in the widely used Gurson–Tvergaard–Needleman (GTN) model. In GTN, coalescence is assumed to occur when the void volume fraction reaches a critical value. Although this critical void volume fraction was initially considered a material constant, it was later shown to depend on factors such as the initial void volume fraction and the applied stress triaxiality [28,46]. Additionally, microvoids can nucleate not only at the onset but also progressively during plastic deformation. Various nucleation models [46,47] have been developed and can be readily incorporated into the GTN framework.

Despite its effectiveness in modeling ductile fracture under high triaxiality conditions, the GTN model lacks the physical fidelity to automatically determine the transition from homogeneous void growth to localized coalescence. This limitation was first highlighted by Thomason [48–50], who argued that an additional localized plastic deformation criterion is necessary to account for void coalescence, emphasizing that localized plastic instability (i.e., the limit load condition) governs the onset of final fracture. Building on this insight, Zhang and co-workers extended the GTN framework by incorporating Thomason's localized plastic instability-based coalescence criterion, leading to the development of the so-called CGM [26–29,51,52]. The CGM captures the full sequence of ductile fracture mechanisms and offers several key advantages over the original GTN model: (1) void nucleation becomes the root cause of ductile failure, while void growth and coalescence are the response of material against loading; (2) for a given loading path, void coalescence in CGM is uniquely determined by the void nucleation parameters, establishing a one-to-one relationship that enables the development of experimental methods for void nucleation parameter identification; (3) the critical void volume fraction is no longer a material constant, reducing the number of model parameters required; and (4) more importantly, the CGM provides a mechanistic foundation to incorporate HE mechanisms such as HELP, HESIV, and HEDE.

The CGM consists of a GTN model and Thomason's localized plastic deformation model for void coalescence. The GTN yield function incorporates the effects of voids on the plastic flow behavior of material and is given by:

$$\phi = \left(\frac{\sigma_{eq}}{\sigma_Y}\right)^2 + 2q_1 f^* \cosh\left(\frac{3q_2\sigma_H}{2\sigma_Y}\right) - 1 - q_3 f^{*2} = 0 \tag{1}$$

where σ_{eq} is the von Mises equivalent stress, σ_h is the hydrostatic stress, σ_Y is the current yield or flow stress of the matrix material, f^* is the effective void volume fraction (defined below) and q_1, q_2 and q_3 are model parameters introduced by Tvergaard to improve the model prediction.

Void evolution is modeled as the sum of growth of the existing void and nucleation of the new voids during the time increment:

$$\dot{f} = \dot{f}_{growth} + \dot{f}_{nucleation} \tag{2}$$

The growth term is driven by the plastic volumetric strain:

$$\dot{f}_{growth} = (1-f)tr(\dot{\boldsymbol{\varepsilon}}_p) \tag{3}$$

where $\boldsymbol{\varepsilon}_p$ is the plastic strain tensor. Void nucleation can be both stress and strain-controlled [30] and a thorough review on void nucleation mechanisms and models for metals can be found in [46]. For the cases considered in this study, we adopt a cluster-type void nucleation assuming voids are nucleated upon the onset of plastic deformation, represented by an initial void volume fraction, f0. In addition to the f0, we employ a widely adopted strain-controlled void nucleation relation where the nucleation rate scales with the plastic strain rate:

$$\dot{f}_{nucleation} = A\dot{\varepsilon}_p \tag{4}$$

where $\dot{\varepsilon}_p$ is the equivalent plastic strain rate and A is a nucleation coefficient.

To account for void coalescence, the GTN model introduces an effective void fraction f^*, defined as:

$$f^* = \begin{cases} f, f \le f_c \\ f_c + \dfrac{f_u - f_c}{f_f - f_c}(f - f_c) \quad , f > f_c \end{cases} \tag{5}$$

Here, f_c denotes the critical void volume fraction that marks the onset of void coalescence, $f_u = \frac{1}{q_1}$ and f_f corresponds to the void volume fraction at final fracture. in the original GTN model, f_c is a pre-selected material constant.

CGM incorporates a localized plastic instability model derived by Thomason as the coalescence criterion. Thomason observed that the sudden transition from a macroscopically homogeneous state of plastic flow to a highly localized internal necking of the intervoid matrix across a single "weakest" sheet of voids coincides with the attainment of the plastic limit-load condition for localized plastic failure of intervoid matrix [53]. The maximum principal stress σ_1 at the onset of plastic localization, $\sigma_1^{Localized}$, is a function of current void geometry. Under plane strain condition and assuming a cylindrical void, the $\sigma_1^{Localized}$, can be calculated as:

$$\frac{\sigma_1^{Localized}}{\sigma_Y} = 0.3\left(\frac{1}{r} - 1\right) + 0.6 \text{with} r = \frac{2}{\sqrt{\exp(\varepsilon_2)}}\left(\frac{3f\exp(\varepsilon_1 + \varepsilon_2)}{4\pi}\right)^{1/3} \tag{6}$$

where ε_1 and ε_2 are the current principal strains [30]. The material point transitions from homogeneous plastic flow to localized plastic flow (start of void coalescence) when:

$$\sigma_1 \ge \sigma_1^{Localized} \tag{7}$$

Thomason has also developed similar coalescence models for 3D loading. Thomason's models were found surprisingly accurate. Further developments of the Thomas models can be found by Benzerga, Pardoen and their colleagues [54–56]. Integrating GTN model for homogeneous plastic flow and Thomason models for localized plastic instability criterion leading to the development of CGM, which serves as the foundation upon which hydrogen effects are later introduced in the H-CGM+ predictive framework.

2.2. Incorporating hydrogen transport

Hydrogen transport in metallic materials is governed by two mechanisms: lattice diffusion and trapping. In the present study, a chemical potential-based formulation is employed to model hydrogen diffusion, following the approach proposed by Di Leo and Anand [57] and implemented as an explicit hydrogen transport framework by Díaz et al. [33]. This formulation is particularly well suited for explicit simulations, as it eliminates the need to calculate hydrostatic stress gradients, which, while not exclusive to implicit solvers, are more readily accessible in that context due to the simpler subroutine calling sequence. In explicit solvers like Abaqus/Explicit, accessing neighboring integration points for gradient calculation requires costly data handling strategies. Moreover, the chemical potential formulation ensures thermodynamic consistency not only in the hydrogen flux but also in the boundary conditions governing uptake [58].

For low-solubility alloys, the occupancy of lattice sites $\theta_L = C_L/N_L$ remains much less than unity, allowing simplifications of the governing equations. In the absence of trapping, hydrogen diffuses through interstitial lattice sites. The concentration in lattice sites C_L, can be determined from the chemical potential, which is the solved dependent variable that drives diffusion. The hydrostatic stress reduces the chemical potential and thus it must be considered in the relationship between C_L and μ_L:

$$C_L = N_L \exp\left(\frac{\mu_L - \mu_L^0 + \overline{V}_H\,\sigma_H}{RT}\right) \tag{8}$$

Here, N_L is the density of available lattice sites, μ_L^0 is a reference chemical potential taken from literature [57], $\overline{V}_H$ is the partial molar volume of hydrogen, and R is the universal gas constant. This expression captures the effects of stress-assisted diffusion or stress-driven diffusion, which has been extensively discussed in the works of Sofronis and McMeeking and confirmed by multiple experimental observations [1].

The hydrogen flux in direction i, $J_L(i)$, is governed by:

$$J_L(i) = -D_L^{\tau}\frac{C_L}{RT}\frac{d\mu_L}{dx_i} \tag{9}$$

where D_L^{τ} is the scaled diffusivity, accounting for time-rescaling between experimental loading (t_{load}) and simulation duration (τ_{step}), i.e., $D_L^{\tau} = D_L t_{load}/\tau_{step}$. This scaling is introduced for numerical convenience, allowing the use of a fixed simulation time (τ_{step}) while capturing the hydrogen diffusion response corresponding to a given physical loading duration (t_{load}). It does not directly simulate long-term exposure but enables normalization of the time-dependent transport problem.

In practice, the evolution of hydrogen concentration is significantly affected by the interaction between hydrogen atoms and microstructural defects. These defects, such as dislocations, grain boundaries, voids, and precipitates, act as trapping sites. The total hydrogen concentration is thus partitioned between lattice (C_L) and trapped (C_T) populations. The local density of available traps N_T increases with plastic deformation due to dislocation generation and is modeled here using several empirical relationships (Fig. 4). Three widely cited trap density–strain

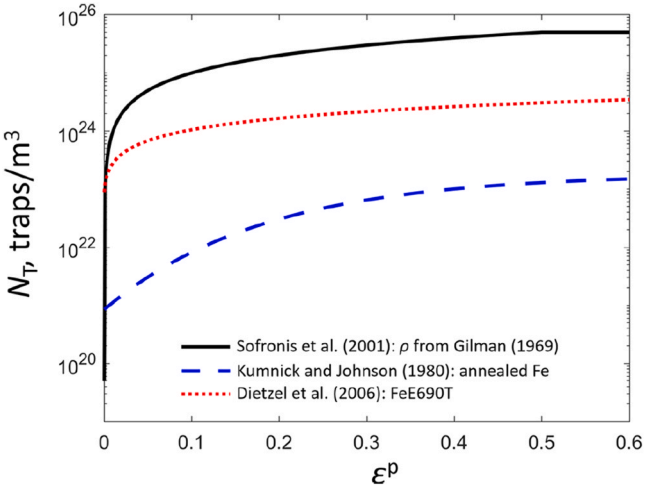


Fig. 4. Trap density as a function of plastic strain for iron and steel for the three empirical relations. This figure is replotted from [1].

relations are considered in this study, here the trap density N_T is expressed in $traps/m^3$:

Dietzel et al. [59], used as the default model:

$$N_T = 8.8 \bullet 10^{22} + 4.8 \bullet 10^{24} {\varepsilon_p}^{0.7} \tag{10}$$

Kumnick and Johnson [31], representing a lower trapping density:

$$\log(N_T) = 23.26 - 2.33\exp\left(-5.5\varepsilon_p\right) \tag{11}$$

Sofronis et al. [60], offering a physically motivated alternative that includes the lattice parameter $L = 2.86\mathring{A}$. This expression is based on dislocation densities and is valid for bcc steels. After a given amount of plastic strain, the trap density becomes independent of the plastic strain:

$$N_T = \begin{cases} \sqrt{2}\,\dfrac{10^{16}}{L}\left(10^4 + 1.2 \bullet 10^{10}\varepsilon_p\right), & \varepsilon_p < 0.5 \\ \sqrt{2}\,\dfrac{10^{16}}{L}\left(10^4 + 6 \bullet 10^{9}\right), & \varepsilon_p \geq 0.5 \end{cases} \tag{12}$$

These expressions reflect the experimentally observed variability in trapping behavior and enable sensitivity studies on how HE depends on plastic deformation history and microstructural features.

Under Oriani's local equilibrium assumption, the occupancy of trap sites θ_T is related to the lattice occupancy θ_L by:

$$\theta_T = \frac{\theta_L K_T}{1 + \theta_L K_T} \tag{13}$$

where $K_T = \exp(E_B/RT)$ is the equilibrium constant associated with the trap binding energy E_B. Using this relationship, the trapped hydrogen concentration C_T can be expressed in terms of the local lattice concentration C_L and the densities of lattice and trap sites, N_L and N_T, as:

$$C_T = \frac{N_T}{1 + \frac{N_L}{K_T C_L}} \tag{14}$$

This equation highlights that, at low lattice hydrogen concentration C_L, most traps remain empty, while at high C_L or for large trap binding energies, C_T approaches the total trap density N_T. This approach captures reversible trapping effects without the need for explicit kinetic modeling, which is computationally expensive and often unnecessary for low-temperature applications. While Oriani's equilibrium assumption [61,62] is widely used, its validity has been extensively examined in the literature. Studies have shown that for low strain rates and quasi-static loading conditions, such as those relevant to pipeline service, the equilibrium approximation remains appropriate. In contrast, for high strain-rate or cyclic loading, the full McNabb-Foster kinetic formulation may become necessary [63,64].

In the present implementation, the plastic-strain-dependent trap density is associated with dislocation-related trapping. This choice does not imply that all traps in the material are assumed to be dislocation-based. Other trapping sites may also contribute to hydrogen storage and embrittlement. For the sake of simplicity and given the generic nature of the present work, these contributions are not explicitly considered in the present study. They are therefore not explicitly introduced as separate evolving trap populations in the present study. Instead, their possible effects are included implicitly through the effective initial hydrogen state, the chosen trap density relation, and the material dependent damage and decohesion parameters.

The total local hydrogen concentration is then obtained as the sum of the lattice and trapped contributions, and represents in this study the effective hydrogen available to influence the material properties within the H-CGM+ framework:

$$C_H = C_L + C_T \tag{15}$$

The resulting hydrogen field reflects the coupling between mechanical fields (stress, strain), hydrogen diffusion and trapping. It governs the local embrittlement behavior through the concentration C_H, which is used in the subsequent sections to affect void nucleation and fracture criteria in the H-CGM+ framework. The assumption that total hydrogen controls fracture and damage is debated in literature. While some studies suggest that only the diffusible (lattice) hydrogen, actively contributes to embrittlement processes, particularly through mechanisms like HELP and HEDE, others argue that traps with low binding energy can dynamically release hydrogen, making C_T relevant under certain conditions. In contrast, strongly binding traps may even mitigate embrittlement by sequestering hydrogen away from critical fracture sites.

2.3. Effect of hydrogen on ductile fracture – hydrogen enhanced void nucleation

For bcc metals, ductile fracture under low hydrogen concentration remains governed by MVC mechanism. However, notable microstructural changes are observed: the size of the highly plastically deformed zone near the fracture surface is reduced, dimples become shallower, and their characteristic size is refined [65]. Over the past decades, numerous numerical studies have been conducted to investigate the influence of hydrogen on ductile fracture behavior [60,66–69]. A central assumption in these studies is that the matrix material within a representative volume element (with or without voids) exhibits significant softening due to the HELP mechanism. This assumption leads to the conclusion that hydrogen promotes earlier plastic localization, both in the presence and absence of voids, as a consequence of matrix softening. Specifically, Yu [70] demonstrated that void growth is accelerated under conditions of pronounced matrix softening. However, the influence of such softening on void coalescence remains limited unless an unrealistically high degree of softening is introduced. Depicting HELP mechanism through matrix softening at the continuum level remains questionable. To date, no conclusive experimental evidence confirms global matrix softening under hydrogen, nor a quantitative relationship between the extent of softening and ductile fracture has been established.

High-resolution X-ray computed tomography has been employed to investigate hydrogen-induced damage in X70 pipeline steel, revealing that hydrogen significantly accelerates void nucleation [71]. While the distinction between void nucleation and growth remains inherently uncertain in such analyses, the study suggests that hydrogen does not substantially accelerate void growth, or at the very least, its effect on growth is far less pronounced than on nucleation. Complementary statistical analysis of HE mechanisms in medium-Mn steels, consisting of a ferritic matrix with embedded austenite, further supports this view. In hydrogen-charged specimens, void nucleation rate was observed to increase by up to a factor of approximately $\times$ 13 compared to uncharged counterparts [72], strongly indicating a dramatic enhancement in void nucleation due to hydrogen.

Motivated by these observations, we adopt a unifying perspective in which the influence of hydrogen on ductile fracture is represented through an additional hydrogen-enhanced void nucleation (HEVN) contribution, superimposed on the baseline nucleation rate observed in air. Rather than treating individual mechanisms separately, the combined effects of HELP and HESIV are interpreted as an enhanced supply of nucleation sites driven by hydrogen–defect interactions. In this framework, HEVN therefore represents their common continuum-level consequence, which manifests as an increased probability of void nucleation, while HELP and HESIV remain distinct microscopic mechanisms. Specifically, HELP facilitates void nucleation by enhancing dislocation activity and weakening particle–matrix interfaces, while HESIV contributes through the formation and stabilization of vacancy clusters [65] that can evolve into nanoscale voids, as illustrated in Fig. 5.

More generally, HEVN may also incorporate contributions from hydrogen-assisted decohesion occurring at particle–matrix interfaces, provided that such processes manifest as void nucleation events. This should be distinguished from the hydrogen-assisted brittle fracture

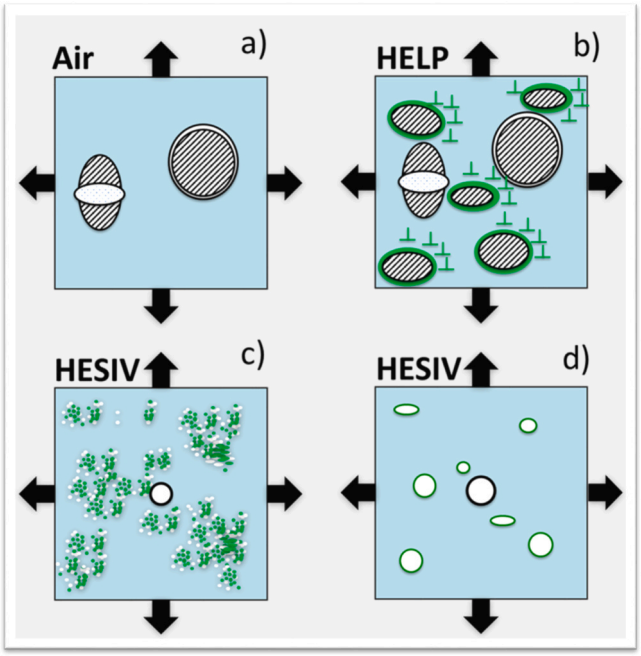


Fig. 5. Schematic illustration of void nucleation mechanisms in air and under hydrogen charging. (a) In air, voids nucleate primarily by particle fracture or particle–matrix interface decohesion. Large particles tend to nucleate voids at early stages of plastic deformation (referred to as primary voids), whereas smaller inclusions typically form secondary voids at higher strains. (b) Under hydrogen charging, hydrogen-enhanced localized plasticity (HELP) promotes dislocation activity and strain localization, leading to the activation of additional particle-related void nucleation sites. (c) Hydrogen-enhanced strain-induced vacancies (HESIV) form during plastic deformation, resulting in the accumulation and stabilization of vacancy clusters. (d) These hydrogen-stabilized vacancy clusters can evolve into nanoscale voids, providing an additional mechanism for void nucleation independent of second-phase particles. Void sizes are not drawn to scale.

mechanism introduced in Section 2.4, which describes decohesion at larger length scales (e.g., along grain boundaries or cleavage planes) leading to abrupt fracture. Within the present formulation, HEVN remains a distributed damage mechanism governing ductile fracture degradation.

Importantly, the HEVN concept is mechanism-informed, although its present quantitative implementation remains phenomenological. It represents a microstructure-dependent measure of the efficiency with which hydrogen–defect interactions are converted into void nucleation events. It therefore reflects underlying material features such as inclusion characteristics, dislocation density evolution, and vacancy formation propensity, providing a physically grounded link between microstructure, hydrogen concentration, and damage evolution.

Based on the above hypothesis, void nucleation is augmented by local hydrogen concentration through a scaling factor. Specifically, the void volume fraction f evolves according to:

$$\dot{f} = \dot{f}_{growth} + (1 + k_n C_H)\dot{f}_{nucleation} \tag{16}$$

where C_H is the local hydrogen concentration, and k_n is a microstructure dependent coefficient that quantifies the influence (sensitivity) of hydrogen on void nucleation, respectively. The HELP mechanism, in which hydrogen enhances dislocation mobility and thus accelerates plastic dilatation. The HESIV, where hydrogen promotes vacancy formation and facilitates void nucleation [28]. This coupling results in locally accelerated damage evolution in hydrogen-rich regions, consistent with experimental observations of reduced ductility and suppressed ductile fracture under hydrogen exposure [17,28]. In the current formulation, k_n scales linearly with C_H. This choice should be regarded as a mechanism-informed phenomenological approximation rather than a first-principles nucleation law. It reflects the physical assumption that hydrogen-assisted mechanisms increase the probability of void nucleation, while the exact quantitative relation still requires calibration from atomistic-scale simulations, micromechanical modelling and experiments.

2.4. Incorporating hydrogen-assisted brittle fracture: H-CGM+

We intend to expand the H-CGM framework to include both the hydrogen-suppressed ductile fracture and hydrogen-assisted brittle fracture and allow these two failure mechanisms to compete under one unified framework. To capture the ductile-to-brittle transition observed in HE, the H-CGM+ model integrates a brittle fracture criterion inspired by the HEDE mechanism. In the present framework, this brittle fracture is assumed to occur at a larger length scale within the matrix, such as along grain boundaries, cleavage planes, or phase boundaries, rather than at particle–matrix interfaces, whose contribution is incorporated in Section 2.3 through HEVN. In particular, brittle fracture may emerge in regions between nucleated voids, where hydrogen accumulation and stress localization promote decohesion of the surrounding matrix. Brittle fracture is often controlled by the maximum principal stress, σ_1. It does not happen in air because the stress needed for direct decohesion of atomic planes is approximately one tenth of elastic modulus [73] which greatly exceeds the applied stress. However, the stress for decohesion can be drastically lowered when exposed to hydrogen. A degradation law originated from quantum mechanical calculations [74,75] that links cohesive strength to hydrogen coverage θ at potential fracture interfaces has been widely used in the literature [17] and is also used in this study. The local hydrogen coverage is computed using a Langmuir-type isotherm:

$$\theta = \frac{c_H}{c_H + \exp(-\Delta g_b^0 / RT)} \tag{17}$$

where c_H is the impurity fraction (molar ratio) calculated as $c_H = C_H[mol/m^3]/N_{Fe}$, with N_{Fe} the molar density of iron, $\Delta g_b^0 = 30kJ/mol$ is the Gibbs free energy difference between the bulk and the fracture interface, and R and T are the universal gas constant and absolute temperature, respectively. This formulation captures the saturation behavior of interfacial hydrogen adsorption. Although the segregation model is applied to the total hydrogen concentration, it should be noted that the hydrogen population governing interfacial segregation remains an open question. Previous studies suggest that only the lattice hydrogen contributes to interfacial segregation because it represents the mobile hydrogen available for segregation. One can also argue that the reversible trapped hydrogen may also contribute [7]. The present formulation therefore adopts the total hydrogen concentration as a practical effective quantity while explicitly acknowledging this uncertainty.

The decohesion threshold under hydrogen influence, γ_{CH}, is modeled using the degradation law originally derived from DFT-based calculations of the H/Fe interface:

$$\gamma_{CH} = \left(1 - 1.0467 \quad \theta + 0.1687 \quad \theta^2\right)\gamma \tag{18}$$

Where γ is a dimensionless material parameter accounting for the cohesive strength in air. This quadratic degradation function, validated across several pipeline and structural steels [13,31], reflects the loss of atomic bonding strength due to interfacial hydrogen accumulation.

Although the Thomason void coalescence criterion and the above hydrogen-assisted brittle fracture criterion are physically distinct, both involve the maximum principal stress, as the relevant mechanical quantity. This common stress measure allows the brittle criterion to be conveniently implemented together with the ductile coalescence criterion in H-CGM, leading to the H-CGM+ framework, which covers the full

spectrum of HE. In the H-CGM+ implementation, brittle fracture is considered to occur when the local maximum principal stress σ_1 exceeds the hydrogen-degraded cohesive threshold:

$$\sigma_1 \geq \gamma_{CH}\sigma_0 \tag{19}$$

This formulation enables calibration of the material's intrinsic resistance to decohesion via the dimensionless parameter γ, and modulates the ductile-to-brittle transition threshold as a function of hydrogen concentration.

To interpret the competition between ductile and brittle fracture mechanisms, the model compares four critical quantities as functions of accumulated plastic strain ε_p:

The maximum principal stress normalized by the flow stress σ_1/σ_Y is compared with the localized coalescence threshold $\sigma_1^{Localized}/\sigma_Y$.

The maximum principal stress normalized by the initial yield stress σ_1/σ_0 is compared with the decohesion threshold γ_{CH}.

A schematic of these competing criteria is shown in Fig. 6. The ratios σ_1/σ_Y and σ_1/σ_0 are increasing and then decreasing functions of the plastic strain. In air, the decohesion threshold γ is too high to intersect with the function σ_1/σ_0, however $\sigma_1^{Localized}/\sigma_Y$ decreases faster than the ratio σ_1/σ_Y and their intersection triggers void coalescence. In the presence of hydrogen, HELP and HESIV accelerate the decrease of $\sigma_1^{Localized}$, leading to a smaller value of plastic strain at void coalescence as shown in Fig. 6(b). In addition, hydrogen decreases the brittle threshold γ_{CH} until a point where it will intersect with σ_1/σ_0, leading to brittle decohesion via HEDE. In a first time the introduction of hydrogen lowers the material ductility by HELP and HESIV until the hydrogen concentration is significant enough to trigger HEDE. It is important to notice on Fig. 6(c) that the transition of failure mode from ductile to brittle implies a sudden shift of the plastic strain $\varepsilon_p^{ductile}$ to $\varepsilon_p^{brittle}$. The loss of ductility is abrupt since the ductile and brittle threshold are compared to two different ratios.

The parameter γ acts as a material-specific resistance to brittle failure: higher values delay the onset of HEDE-dominated failure, while lower values make the material more sensitive to interfacial decohesion. This framework allows for a seamless transition between ductile, brittle, and mixed fracture modes as a function of local hydrogen concentration, constraint, and plastic deformation history.

3. Predictability of H-CGM+ and Hydrogen-constraint interaction

H-CGM+ is a micromechanically informed predictive framework that treats hydrogen-enhanced void nucleation (HEVN) as the root cause of hydrogen-suppressed ductile fracture. By integrating both ductile and brittle fracture mechanisms within a unified formulation, the framework establishes a direct link between fracture behavior and the underlying microstructure, local hydrogen concentration, and loading conditions. In addition to the standard CGM parameters, the model introduces two microstructure-dependent parameters: k_n, representing the hydrogen-enhanced void nucleation rate, and γ, representing the dimensionless cohesive strength associated with decohesion. The objective of this section is to demonstrate the predictive capability of H-CGM+, with particular focus on the synergistic effects of hydrogen and crack-tip constraint in SENT specimens. Given that most available experimental studies do not provide sufficient data for a complete calibration of both constitutive and damage parameters, a generic BCC steel dataset is adopted. This allows for a systematic investigation of the governing mechanisms and their interactions, while isolating the roles of hydrogen

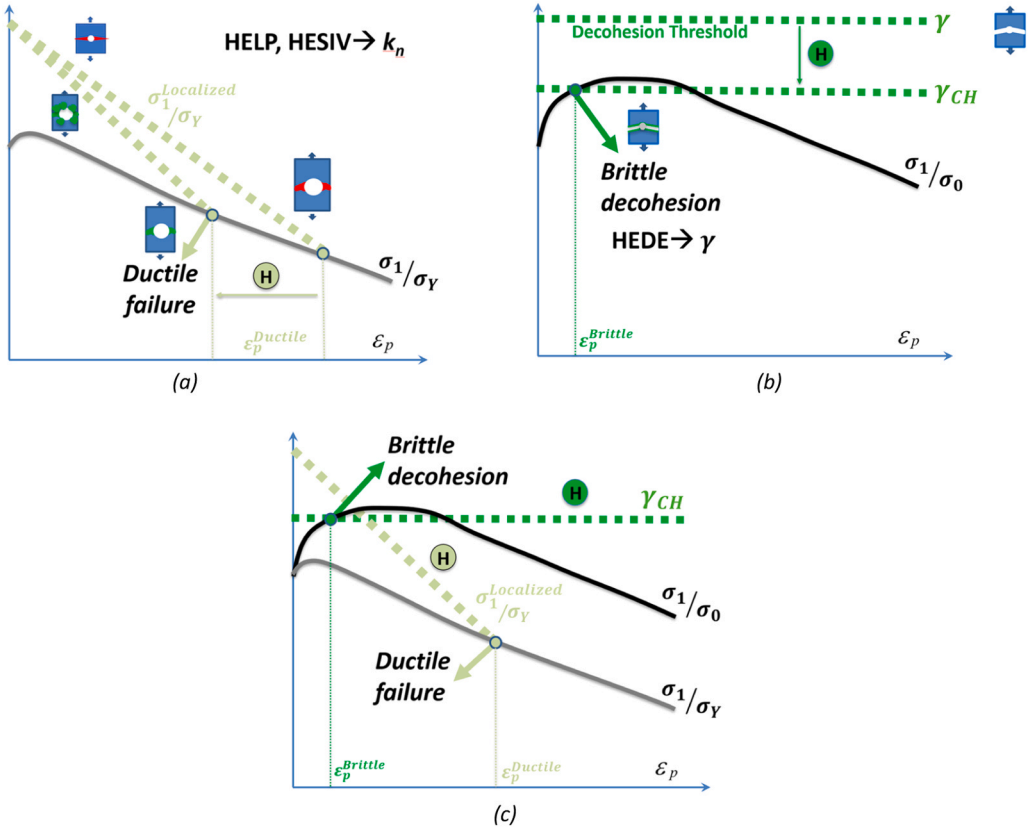


Fig. 6. Schematic of (a) the ductile fracture mechanism, (b) the brittle decohesion mechanism and (c) the competition between stress-controlled decohesion and ductile failure modified by hydrogen. In the presence of hydrogen, two scenarios are considered. For low hydrogen concentrations, γ_{CH} remains too high and decohesion does not occur, but via HELP and HESIV hydrogen can cause a ductility loss, void coalescence occurs and $\varepsilon_p^{ductile}$ is reduced. For high hydrogen concentrations, γ_{CH} and σ_1/σ_0 crosses at $\varepsilon_p^{brittle}$ before reaching $\varepsilon_p^{ductile}$. Brittle decohesion occurs via HEDE and the plastic strain at failure is significantly reduced.

concentration, constraint level, and material sensitivity in controlling fracture behavior.

3.1. Experimental evidence of fracture toughness under hydrogen

Hydrogen is known to reduce ductility and fracture toughness and to accelerate fatigue crack growth. Compared with ductility and fatigue behavior, however, the effect of hydrogen on fracture toughness remains much less well understood. In this study, we therefore focus specifically on fracture toughness.

It is generally accepted that hydrogen deteriorates the fracture resistance of metallic alloys, with higher hydrogen concentration leading to a greater reduction in fracture toughness [76]. However, a unified methodology for measuring fracture toughness under hydrogen is still lacking. The measured fracture toughness depends strongly on specimen geometry, loading mode, and loading rate, and is typically reported in terms of parameters such as the stress intensity factor K, the J-integral, or CTOD. The influence of specimen geometry and loading mode is commonly referred to as crack-tip constraint. As discussed in the introduction, several scenarios have been proposed regarding the effect of constraint on fracture toughness under hydrogen exposure, which is further complicated by the level of hydrogen concentration.

Halilović et al. investigated AISI 4340 steel under hydrogen charging using specimens with different constraint levels [77]. Their experimental results for fracture toughness, expressed as $J_{0.2\,\mathrm{mm}}$, for both high-constraint ($a/W = 0.5$) and low-constraint ($a/W = 0.1$) specimens as a function of loading rate are summarized in Fig. 7, where lower loading rates correspond to higher hydrogen exposure. Several key observations can be made: (1) fracture toughness decreases with increasing hydrogen exposure; (2) at low hydrogen exposure, low-constraint specimens exhibit higher fracture toughness (scenario H1); (3) at high hydrogen exposure, fracture toughness becomes less sensitive to crack-tip constraint (scenario H2); and (4) the reduction in fracture toughness is more pronounced in low-constraint specimens than in high-constraint ones.

More recently, a striking and somewhat counterintuitive trend has been reported [43], where low-constraint specimens exhibit lower fracture toughness than high-constraint ones (scenario H3). This observation has raised significant concern in the offshore industry, and ongoing experimental studies suggest that such behavior may occur in pipeline steels under certain conditions.

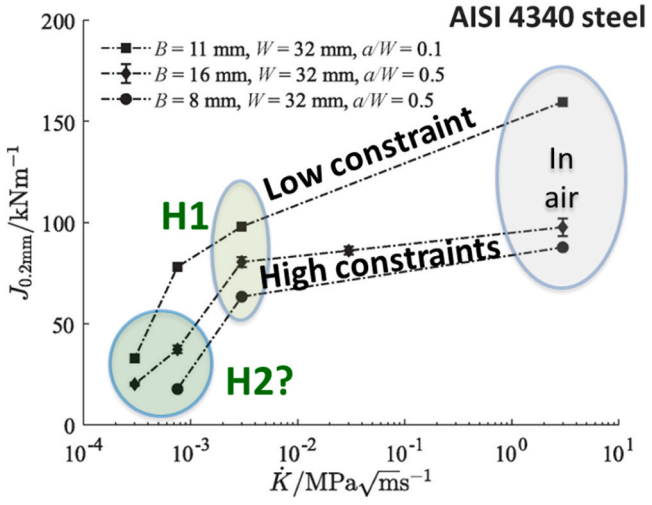


Fig. 7. Experimental results of the combined effect of crack tip constraint and hydrogen loading rate for AISI 4340 steel. Specimen with a/W = 0.1 represents the low constraint specimen while the specimens with a/W = 0.5 represents the high constraint specimen [77].

3.2. Material properties, parameters and simulation setup

3.2.1. Material property data set

A generic high-strength ferritic pipeline steel has been selected as the sample material for the demonstration of the predictability of the H-CGM+. The mechanical response of the pipeline steel investigated is modeled using an isotropic power-law hardening formulation. The flow stress σ_Y as a function of accumulated plastic strain ε_p is given by:

$$\sigma_Y = \sigma_0 \left(1 + \frac{\varepsilon_p E}{\sigma_0}\right)^n \tag{20}$$

where σ_0 is the initial yield stress, E is Young's modulus, and n is the hardening exponent. This formulation provides a smooth transition from elastic to plastic behavior and reflects the moderate work-hardening behavior typical of pipeline steels.

The mechanical parameters used in this study include elastic, plastic, and void damage parameters used in the CGM. Note that the density value is selected to ensure numerical stability of the explicit integration scheme, where the time step is governed by the material's wave speed. A reduced density increases the allowable time increment, enabling efficient quasi-static simulations. The values are summarized in Table 1.

For hydrogen transport, the model parameters adopted correspond to the reported values for hydrogen in BCC steels [1,31,33]. These parameters govern lattice diffusivity, trap binding, and chemical potential under hydrostatic stress. The values used are summarized in Table 2.

3.2.2. Simulation setup

Commercial software ABAQUS with its user subroutine VUMAT and VUMATHT was utilized to implement the H-CGM+ predictive framework.

The fracture behavior of the material is studied through a series of finite element simulations using SENT specimens under plane strain conditions. All simulations are performed using an explicit dynamic solver, and hydrogen diffusion is incorporated through a custom chemical potential-based subroutine, as described in Section 2.2.

The geometry of the SENT specimens employed in this study is illustrated in Fig. 8(a). The specimen width is fixed at $w = 10mm$, and the total length is $2L = 100mm$. To investigate the effect of crack-tip constraint, five different initial crack lengths are considered: $a = 1,\ 2,\ 3,\ 4,\ 5$ mm, corresponding to a/w ratios of 0.1–0.5. The choice of the specimen geometry and crack sizes aims to greatly vary the crack tip constraint and is consistent with the practices without hydrogen [35]. Due to symmetry, only half of the specimen is modeled. All simulations assume quasi-static loading conditions, and inertia effects are verified to be negligible.

The mesh configuration is shown in Fig. 8(b). The mesh is refined in the crack-tip region, where high gradients in stress, strain, and hydrogen

Table 1
Mechanical properties of the generic pipeline steel.

Young's Modulus	E	210000	MPa	Elastic stiffness
Poisson's ratio	ν	0.3		Elastic response
Initial yield stress	σ_0	450	MPa	Onset of plasticity
Hardening exponent	n	0.1		Work-hardening behavior
Tvergaard coefficients	$q_1,\ q_2,\ q_3$	1.5, 1.0, 2.25 -		Void interaction modifiers
Initial void volume fraction	f_0	0.00015		Pre-existing porosity
Final void volume fraction	f_F	0.15		Fracture porosity limit
Nucleation amplitude	A	0.016		Void nucleation rate in air
Nucleation magnifier	k_n	[0,1]		H-accelerated nucleation
Numerical density	d	$1e-7$	kg/mm^3	For explicit integration

Table 2
Hydrogen diffusion and trapping parameters.

Intrinsic lattice diffusivity	D_L^0	0.012	mm^2/s	At reference temperature
Activation energy for diffusion	E_A	9.95	kJ/mol	Arrhenius-type behavior
Lattice site density	N_L	$5.1e+29$	$sites/m^3$	Available interstitial sites
Partial molar volume of H	$\overline{V}_H$	2000	mm^3/mol	Volume change per H atom
Reference chemical potential	μ_L^0	28.6	kJ/mol	For lattice hydrogen
Trap binding energy	E_B	27.6	kJ/mol	Dislocation trap strength
Temperature	T	300	K	Ambient temperature

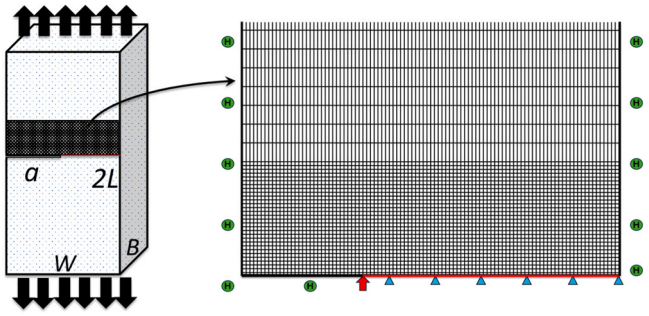


Fig. 8. (a) Schematic of the SENT specimen and (b) crack tip mesh arrangement. Only half of the specimen is considered thanks to the symmetry along the crack ligament. The symmetrical boundary condition is represented by blue triangles; the crack tip is pointed by the red arrow and the chemical potential boundary condition at the specimen's surface is symbolized by the green circles.

concentration are expected. The effect of mesh size has been well studied in literature and in this study we fix the mesh size to be 0.1 mm. A uniform element size of $0.1 \times 0.1mm$ is used in the near-tip region, with the first row along the crack plane refined to $0.1 \times 0.05mm$ to accurately resolve damage initiation [78,79]. The element type used is CPE4RT (4-node, bi-linear, temperature-displacement, plane strain with reduced integration) for simulations including hydrogen transport. For purely mechanical analyses without hydrogen, the CPE4R element is used instead, as no thermal (chemical potential) degree of freedom is needed.

A displacement-controlled loading is applied at the top of the specimen, with a total vertical displacement of 1 mm imposed over a simulation time of $\tau_{step} = 1$. To ensure physically realistic hydrogen distribution, the lattice diffusivity is scaled according to $D_L^{\tau} = D_L t_{load}/\tau_{step}$ where $t_{load} = 20\,hours$ represents the physical loading duration. This time-scaling approach has been validated in previous work [33] and ensures that the hydrogen distribution at the end of the simulation reflects quasi-static diffusion conditions.

This setup enables consistent evaluation of fracture resistance across varying levels of geometric constraint and hydrogen content, while maintaining control over numerical stability and diffusion-reaction coupling. The subsequent sections analyze the results in terms of CTOD, failure mode transition, and sensitivity to embrittlement parameters.

3.3. Predictability of H-CGM+

The predictive capability of the framework is demonstrated through a series of parametric studies designed to isolate the effects of key embrittlement mechanisms, including HELP and HESIV, and HEDE. The role of plastic strain–dependent hydrogen trapping is evaluated by comparing three established trapping models. The analysis then extends to structural and environmental factors, starting with the influence of geometric constraint under both air and hydrogen conditions, followed by the effect of initial lattice hydrogen concentration on fracture behavior. Finally, the coupled effects of constraint and hydrogen embrittlement are examined to elucidate their influence on ductile-to-brittle transition behavior. Consistent with the case in air, CTOD is adopted as the primary measure of fracture resistance, while crack-tip constraint is directly characterized by the a/W ratio in SENT specimens, rather than by T- or Q-stress. This framework enables a quantitative assessment of the competition between ductile and brittle fracture modes over a range of hydrogen concentrations, constraint levels, and material sensitivities.

3.3.1. Effect of hydrogen accelerated void nucleation, k_n

We first examine the influence of the parameter k_n, which controls the sensitivity of void nucleation to local hydrogen concentration and thereby directly links the HELP and HESIV mechanisms (Fig. 5). Simulations are performed on a SENT specimen with moderate constraint level, $a/w = 0.3$, and the CTOD is evaluated at crack propagation length $\Delta a = 0.1mm$. The results are shown in Fig. 9.

The reference case with $k_n = 0$ (blue curve) represents a model where hydrogen influences fracture only via the HEDE mechanism. Hydrogen does not accelerate void nucleation, but the baseline CGM void nucleation and growth processes remain active. Therefore, at low hydrogen concentration, failure remains ductile and is governed by void coalescence. In this case, the CTOD response remains unaffected by hydrogen concentration until a critical threshold is reached. Beyond that threshold, decohesion at the crack tip initiates a brittle fracture, causing a sharp drop in CTOD. This creates a distinct two-plateau behavior: a relatively flat response at low hydrogen concentration followed by a sudden transition to brittle failure at higher concentrations.

As k_n increases, reflecting stronger HELP and HESIV effects, the entire CTOD curve shifts downward. Unlike HEDE, which affects the fracture threshold, HELP and HESIV influence the plastic deformation process itself, accelerating void nucleation and damage accumulation throughout the crack path. Consequently, even before decohesion dominates, CTOD values decrease progressively with increasing hydrogen content. This effect is particularly visible in the second plateau region (post-transition), indicating that HELP and HESIV reduce not only the initiation resistance but also the resistance during crack propagation.

These results demonstrate that the parameter k_n controls how gradual or abrupt the ductile-to-brittle transition appears in the CTOD response. In the absence of HELP and HESIV, the model exhibits a sharp fracture mode switch. When HELP and HESIV are active, this transition is smoother and more distributed, consistent with experimental obser-

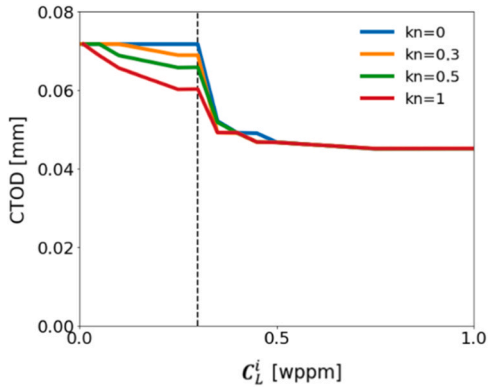


Fig. 9. Fracture toughness (CTOD) versus initial lattice hydrogen concentration C_L^i for $a/w = 0.3$ and $\Delta a = 0.1mm$. The vertical black dashed line marks the highest initial lattice hydrogen concentration for which the fracture of the crack tip element is ductile. On the right side of the dashed line, the crack tip element fails in a brittle manner.

vations of progressive embrittlement. The following section investigates how the HEDE mechanism, governed by the parameter γ, affects the brittle fracture threshold and transition onset.

The narrow transition window in CTOD, which is commonly observed in hydrogen-assisted fracture and often discussed in terms of a critical hydrogen concentration, is naturally captured in the present framework through the micromechanical competition between hydrogen-enhanced void nucleation leading to early coalescence and hydrogen-enhanced decohesion failure. This provides a quantitative description of the ductile-to-brittle transition within H-CGM+. Interestingly, the resulting picture is qualitatively comparable to the transition window envisioned in HELP+HEDE-type synergistic frameworks [76], providing useful consistency between the present micromechanical model and existing model descriptions.

3.3.2. *Effect of hydrogen-assisted decohesion γ*

This section investigates the influence of the parameter γ, which governs the hydrogen-free cohesive strength and thereby controls the susceptibility to decohesion via the HEDE mechanism. Lower values of γ represent materials with intrinsically lower cohesive strength, where brittle fracture is more easily triggered by local hydrogen accumulation. Simulations are conducted for the medium constraint case (a/w $= 0.3$), with CTOD evaluated at crack extensions $\Delta a = 0.1mm$ and $\Delta a = 0.5mm$. The results are shown in Fig. 10.

In the limiting case where $\gamma \rightarrow +\infty$ (red curves), decohesion is effectively disabled, and fracture is governed solely by ductile damage mechanisms, including HELP and HESIV (with $k_n > 0$). The resulting CTOD response shows a gradual and continuous decrease with increasing hydrogen concentration, but no abrupt transition, since the HEDE failure mode is not active.

As γ decreases, the model becomes increasingly sensitive to hydrogen coverage at fracture-critical interfaces. Once the local hydrogen concentration is sufficient to reduce the cohesive threshold below the principal stress, brittle fracture initiates. This is reflected in the CTOD curves by the emergence of a sharp drop at intermediate hydrogen concentrations, signaling the onset of decohesion-driven failure. The location of this transition shifts to lower hydrogen levels as γ is further reduced, showing that more embrittlement-prone materials require less hydrogen to initiate brittle failure.

The curves also reveal that, while the HELP and HESIV mechanisms affect the overall resistance (both plateaus), the HEDE mechanism introduces a nonlinearity in the response by sharply lowering CTOD in a narrow concentration range. The interaction between these mechanisms can therefore be tuned via $k_n$and γ to reproduce material-specific fracture behavior, including abrupt transitions, gradual degradation, or mixed-mode responses.

Comparing subplots (a) and (b) in Fig. 10, the transition to brittle fracture appears at slightly higher hydrogen levels when CTOD is evaluated further along the crack path ($\Delta a = 0.5mm$). This indicates that, for moderate hydrogen concentrations, brittle fracture may initiate locally but ductile propagation can persist behind the crack tip. As hydrogen increases, the brittle zone extends inward, reducing CTOD even at larger propagation lengths.

These results confirm that the parameter γ controls the hydrogen threshold for initiating a brittle failure. In combination with k_n, it enables calibration of the ductile-to-brittle transition location and steepness, making the H-CGM+ model adaptable to a wide range of embrittlement behaviors. However, as noted in the next section, neither parameter controls the amplitude of the CTOD drop at the transition, which is also influenced by the local hydrogen distribution and material trapping characteristics.

3.3.3. *Effect of trapping model on crack tip hydrogen distribution*

To assess the role of hydrogen trapping on embrittlement behavior, simulations are performed using three distinct empirical relationships that relate trap density N_T to plastic strain ε_p: Kumnick and Johnson [31], Dietzel et al. [1], and Sofronis et al. [60]. These models represent increasing trapping intensities and reflect microstructural differences such as dislocation density evolution. The objective is to evaluate how these models affect local hydrogen distribution near the crack tip and the resulting fracture resistance.

Fig. 11 presents the hydrogen concentration fields near the crack tip at the same instant, shortly before the onset of crack propagation, for the medium constraint geometry (a/w $= 0.3$) with an initial lattice hydrogen concentration of 0.5 wppm. The results show that models with higher trap densities (e.g., Sofronis) lead to significantly more localized hydrogen accumulation at the crack tip. This is due to the high plastic strain in this region, which, under strain-dependent trapping relations, produces a locally dense network of hydrogen traps.

The spatial control mechanism varies between models: for low trap density (Kumnick and Johnson), hydrogen follows hydrostatic stress gradients, spreading more diffusely throughout the ligament. In contrast, high-density models (e.g., Sofronis) shift the control to plastic deformation, localizing hydrogen sharply at the crack tip where strain is concentrated.

Despite these differences, the CTOD resistance curves shown in Fig. 12 reveal only modest variations between the trapping models, particularly at early crack propagation stages. The onset of

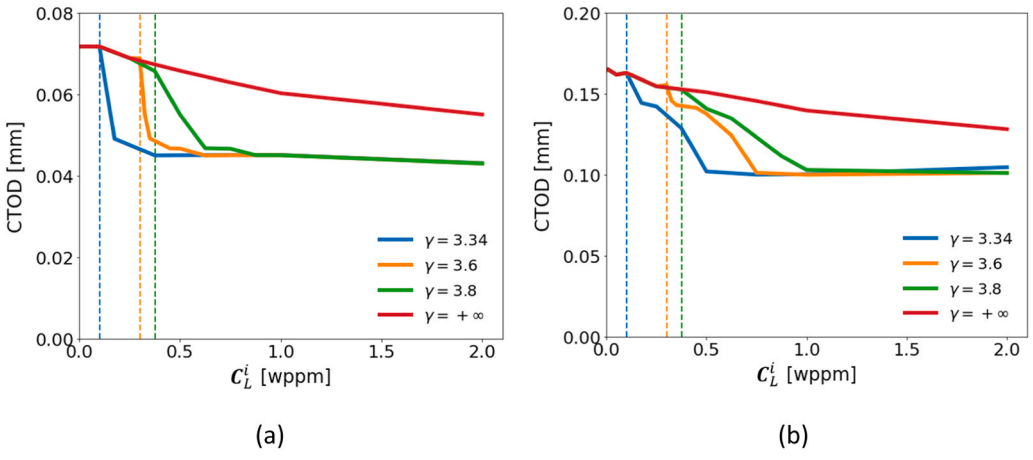


Fig. 10. CTOD versus initial lattice hydrogen concentration C_L^i for a/w = 0.3 and (a) Δa = 0.1mm, (b) Δa = 0.5mm. The vertical dashed lines mark the highest initial lattice hydrogen concentration for which the crack tip element fracture is ductile. Each dashed line is meaningful for the corresponding-colored curve. In the case of the red curve ($\gamma = +\infty$), the fracture is always ductile, the brittle threshold is deactivated/unreachable. The lowering of the HEDE-controlling parameter γ enhances the material sensitivity to hydrogen-promoted decohesion. The critical hydrogen concentration at which the ductile to brittle transition occurs is decreased.

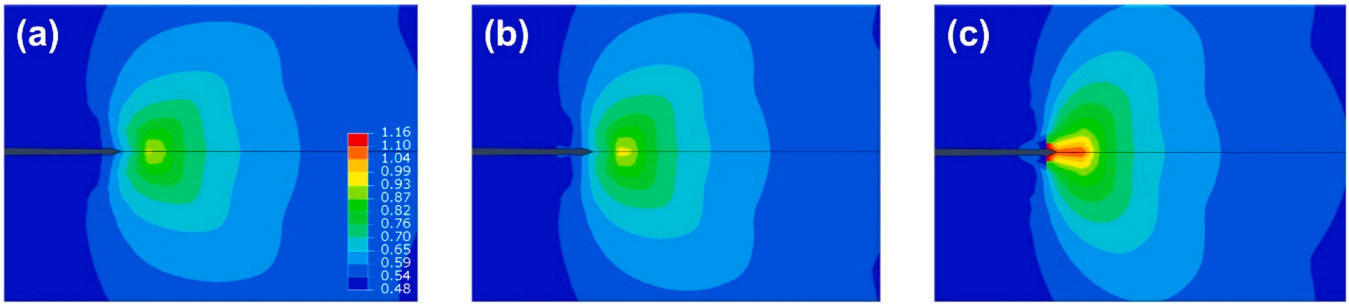


Fig. 11. Local total hydrogen distribution at the crack tip for the three different trapping models shown in Fig. 4. (a) Kumnick and Johnson Eq. 11, (b) Dietzel et al. Eq. 10, (c) Sofronis et al. Eq. 12. The trap density in (c) exceeds the trap density in (b), which in turn is higher than in (a). The trap density influences both the hydrogen concentration values and the hydrogen distribution. A higher density of traps leads to more hydrogen in the plastic region and shifts the location of the maximum hydrogen concentration towards the crack tip, where the maximum of plastic strain is located.

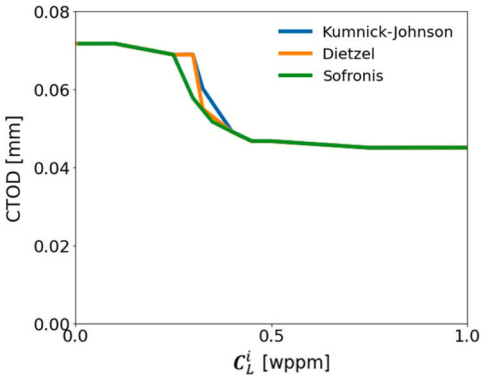


Fig. 12. Resistance curves to HE for Δa = 0.1mm, a/w = 0.3 and the three different trapping models. The three trapping models lead to curves that mostly overlap in the ductile and the brittle regions (i.e. low and high initial lattice hydrogen concentrations). A small reduction of the critical initial lattice hydrogen concentration for the ductile to brittle transition is visible for higher trapping densities. However, despite significant differences in trap densities and hydrogen distribution, the trapping model seems surprisingly to have a low impact on the resistance curve of the material.

embrittlement and the ductile-to-brittle transition occur at similar hydrogen concentrations, suggesting that under the current conditions, the fracture process is not strongly sensitive to the specific trap model. This apparent insensitivity may result from the relatively high initial lattice hydrogen concentration and the low trap binding energy E_B used in the simulations. These values exceed typical lattice solubility values reported in literature and may effectively saturate the traps regardless of the model. As a result, sufficient local hydrogen concentration are reached to activate HELP, HESIV, and HEDE mechanisms in all cases, thereby reducing the influence of trapping details on the simulated fracture behavior.

However, the influence of the trapping model may become more pronounced in situations where plastic strain is higher and more widely distributed, such as in notched round bar (NRB) tests or under global plastic collapse conditions. In such cases, differences in trap evolution could affect the spatial hydrogen profile and shift the onset or propagation of embrittlement. Similarly, the effect may be amplified for lower hydrogen concentration or longer exposure times, where trapping plays a more dominant role in local hydrogen retention.

In summary, while the current SENT specimen simulations indicate limited sensitivity to the trapping model, the framework remains capable of capturing trapping effects and should be re-evaluated under alternative loading scenarios or microstructural conditions.

3.4. Hydrogen-constraint interaction

3.4.1. Effect of geometric constraint in air and hydrogen

The influence of geometric constraint on fracture resistance is first examined by varying the initial crack length a, which modifies the a/w ratio from 0.1 to 0.5. Fig. 13 (a) presents the CTOD evolution as a function of crack propagation in air (i.e. in the absence of hydrogen). As expected from previous studies on SENT specimens [37,39], an increase in initial crack length leads to higher constraint at the crack tip and correspondingly lower CTOD values, indicating reduced ductile crack growth resistance. This trend reflects the well-established relationship between stress triaxiality and fracture toughness: higher constraint increases hydrostatic stress, thereby promoting void coalescence and premature failure.

Fig. 13 (b) shows the corresponding results under hydrogen exposure, with an initial uniform lattice hydrogen concentration of 1 wppm throughout the specimen. The presence of hydrogen significantly reduces the CTOD for all geometries, confirming its detrimental effect on fracture resistance. Furthermore, the fracture behavior transitions from fully ductile in air to mixed-mode under hydrogen.

The effect of geometric constraint remains visible under hydrogen exposure: higher a/w ratios lead to lower CTOD values. However, an important shift is observed in the interaction between constraint and embrittlement. In the presence of hydrogen, specimens with higher constraints exhibit a relatively more pronounced reduction in CTOD, and the distinction between curves corresponding to $a/w = 0.3,\ 0.4,\ 0.5$ becomes less significant. This suggests that, under hydrogen-charged conditions, crack-tip constraint amplifies the embrittlement effect and accelerates the onset of brittle fracture.

These results demonstrate that the H-CGM+ model correctly captures both the constraint-dependent ductile fracture behavior in air and the embrittlement sensitivity introduced by hydrogen. The following sections explore how this trend evolves with varying hydrogen concentration and how it interacts with the underlying embrittlement mechanisms.

3.4.2. Influence of initial hydrogen concentration

To investigate the sensitivity of fracture behavior to hydrogen content, simulations are conducted on SENT specimens with a fixed geometry of $a/w = 0.3$ for a range of initial lattice hydrogen concentration C_L^i. Fig. 14 presents the evolution of CTOD versus normalized crack extension for initial hydrogen contents from 0 to 2 wppm. It should be noted that the value of 2 wppm was selected to represent a severe embrittlement scenario. Under real service conditions, particularly for pipeline steels with low hydrogen solubility, this value likely represents an upper bound on the hydrogen concentration that the material would experience.

At low concentrations ($\leq$ 0.25 wppm), the effect of hydrogen is limited to a modest reduction in CTOD and overall ductility. The

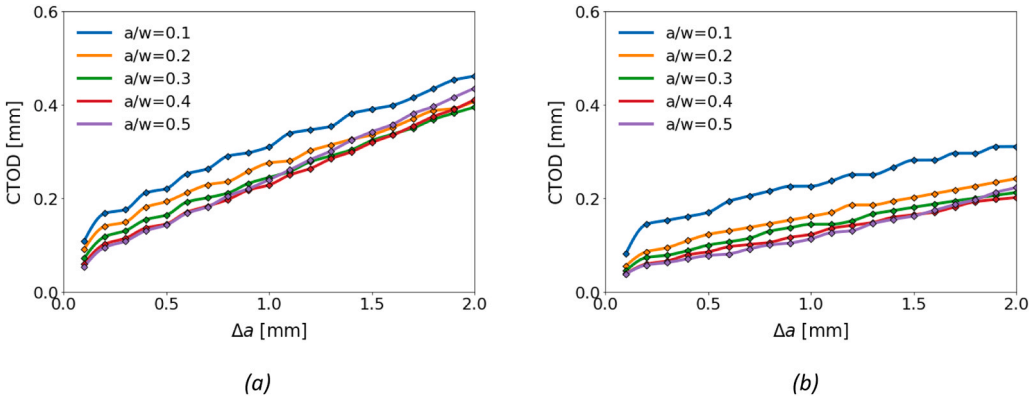


Fig. 13. CTOD against crack propagation length for the 5 different SENT geometries (a) in air, (b) with initial lattice hydrogen concentration of 1 wppm. Subplot (a) confirms that the results follow the accepted fact that in air higher constraints at the crack tip lead to lower resistance to crack propagation. The significant loss of fracture toughness in (b) confirms the detrimental effect of hydrogen for all the five initial crack lengths. Besides, the effect of the crack constraint and on the fracture toughness known in air remains true in hydrogen environments.

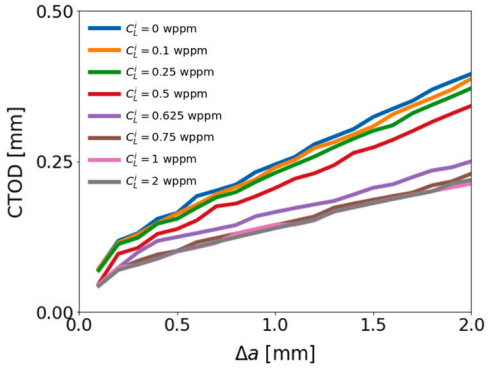


Fig. 14. *CTOD versus crack propagation for SENT with* $a/w = 0.3$. The red curve (0.5 wppm) marks the acceleration of the material's resistance curve deterioration as brittle fracture starts occurring. For lower concentrations, only HELP and HESIV intervene to a small extent. For high hydrogen concentrations, at which the first two millimeters of crack propagation is brittle, the gap between the curves is narrowing.

fracture mode remains fully ductile across the crack path, and the damage evolution resembles that observed in air. As the hydrogen concentration increases, a transition begins to emerge: at 0.75 wppm, brittle failure initiates at the crack tip while the remainder of the crack path still undergoes ductile fracture. This mixed-mode behavior illustrates the spatial variation in local embrittlement, driven by gradients in hydrostatic stress and hydrogen accumulation.

At higher hydrogen levels (e.g., 1.5–2.0 wppm), the brittle mode dominates a larger portion of the crack path, resulting in a sharp drop in CTOD and reduced resistance to crack propagation. The progression from ductile to mixed to brittle failure with increasing hydrogen concentration is captured naturally by H-CGM+, which incorporates both stress-assisted hydrogen transport and hydrogen-sensitive void evolution and decohesion mechanisms.

These results demonstrate that the model can reproduce the gradual embrittlement process observed in experimental studies, where local stress and hydrogen concentration jointly determine the fracture mode. Moreover, the smooth transition between regimes suggests that both HELP and HESIV and HEDE mechanisms are essential to capturing the full range of hydrogen-assisted failure phenomena. The interplay between hydrogen concentration and geometric constraint is explored in more detail in the following section.

3.4.3. Interaction between constraint and HE

To further explore how geometric constraint interacts with HE, CTOD values are compared across the five a/w ratios at crack initiation (i.e. failure of the crack tip element). Fig. 15 presents fracture toughness curves, expressed as CTOD at $\Delta a = 0.1mm$ of crack extension, as a function of constraint and initial lattice hydrogen concentration. Additionally, the ductile or brittle fracture mode is indicated in the figure respectively with stars and triangles. At crack initiation, the trend in CTOD mirrors that observed in air: specimens with higher constraint (larger a/w) exhibit lower fracture resistance, regardless of hydrogen level.

To clarify these observations, Fig. 16 reorganizes the data by plotting CTOD and normalized CTOD at crack initiation ($\Delta a = 0.1mm$) versus initial lattice hydrogen concentration. These plots confirm two important effects. First, the absolute reduction in CTOD increases with both hydrogen concentration and geometric constraint. Second, when normalized by the hydrogen-free values, the embrittlement appears more severe in high-constraint specimens in the range of hydrogen concentration where the ductile-to-brittle transition has not occurred yet for the low-constraint specimens. On the opposite, for high concentrations that make all the specimens reach brittleness, the toughness loss is greater for low-constraint specimens.

These findings highlight the dual role of geometric constraint: it not only reduces fracture toughness in air due to increased triaxiality but also amplifies HE by accelerating the onset of brittle failure, as depicted in Fig. 17. At low hydrogen content, the fracture mode remains ductile, and the constraint effect follows classic void-dominated damage trends.

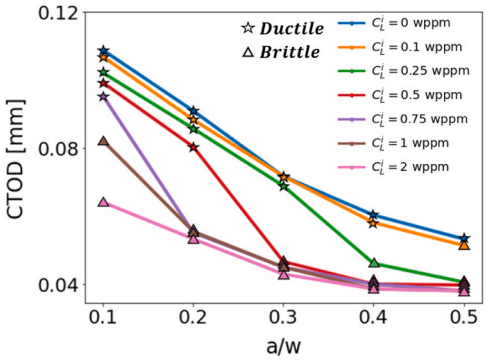


Fig. 15. *CTOD at* $\Delta a = 0.1$ *mm* of crack propagation against the geometrical constraint (a/w) for different initial hydrogen concentrations. Each data point is marked by either a star indicating a ductile fracture mode at crack initiation or by a triangle indicating a brittle fracture mode at the crack initiation. The trend supports the scenarios H1 presented in Fig. 2, at the very start of crack propagation.

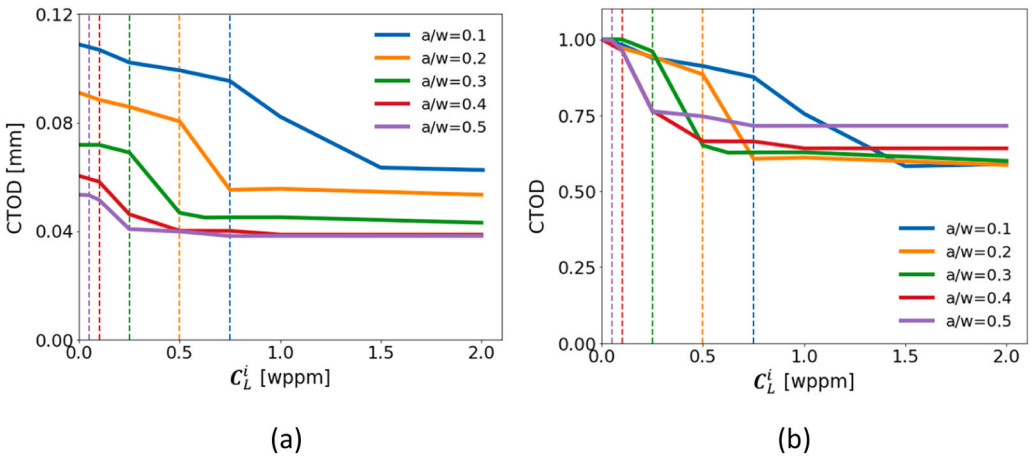


Fig. 16. *(a) CTOD and (b) normalized CTOD at* $\Delta a = 0.1$ *mm* crack propagation versus initial lattice hydrogen concentration C_L^i. The vertical dashed lines mark the highest initial lattice hydrogen concentration for which the crack propagation fracture mode is ductile. Each dashed line is meaningful for the corresponding-colored curve. A higher crack tip constraint results in a lower fracture toughness and a lower critical initial lattice hydrogen concentration for the ductile to brittle transition (a). In addition, according to (b), apart from very low hydrogen content cases or very brittle cases, it induces a higher susceptibility to HE (greater relative drop of the fracture toughness).

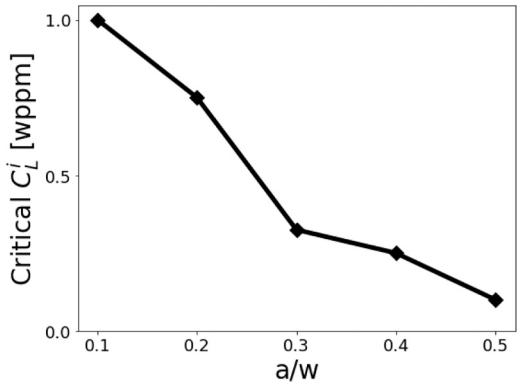


Fig. 17. *Critical* C_L^i, i.e. highest initial lattice hydrogen concentration for which the crack propagation fracture mode is only ductile, against the geometrical constraint parameter (a/w). The constraint lowers the minimal hydrogen concentration necessary to reveal the ductile-to-brittle transition.

At intermediate concentrations, constraint lowers the threshold for mixed-mode and brittle transitions, making constraint more influential. At high hydrogen levels, all geometries tend toward brittle behavior, and the relative influence of constraint becomes less pronounced.

These results support the interpretation that geometric constraint and hydrogen act synergistically to reduce fracture toughness, particularly during the transition regime.

4. Concluding remarks

We have developed H-CGM+, a microstructure-informed, void-based framework for predicting fracture toughness (CTOD) under hydrogen charging. A central contribution of this work is the introduction of hydrogen-enhanced void nucleation (HEVN) as the primary mechanism governing hydrogen-suppressed ductile fracture with reduced dimple sizes. Within this framework, HEVN provides a unifying interpretation of the roles of HELP and HESIV by linking hydrogen–defect interactions to enhanced void nucleation. The model translates the fundamental HELP, HESIV, and HEDE mechanisms into two competing fracture responses—hydrogen-suppressed ductile fracture and hydrogen-assisted brittle fracture, and couples them with stress-assisted diffusion and strain-dependent trapping through a chemical potential-based hydrogen transport formulation. Using SENT specimens, hydrogen–constraint interactions are investigated by coupling mechanical loading and hydrogen transport, which jointly govern damage evolution within a physically based, microstructure-informed framework. The main outcomes of this work, including model developments and mechanistic insights, are summarized below.

Unified predictive framework for competing fracture modes. H-CGM+ embeds hydrogen-suppressed ductile fracture (MVC) and hydrogen-assisted brittle fracture within a single micromechanical formulation, enabling their competition under coupled mechanical–transport fields. By varying the two microstructure-dependent parameters, the HEVN rate (k_n) and the dimensionless cohesive strength (γ) in air, the framework reproduces multiple HE phenomena observed in experiments, including progressive fracture toughness degradation, ductile-to-brittle transitions in CTOD, and mixed-mode fracture behavior within a single specimen.

Distinct mechanistic roles of k_n *and* γ. The parameter k_n associated with HELP- and HESIV-mediated processes, governs hydrogen-induced degradation of ductile fracture resistance by accelerating void nucleation under hydrogen-affected plasticity. The parameter γ, associated with HEDE, controls the threshold for hydrogen-assisted brittle fracture through cohesive degradation. Their combined influence determines whether embrittlement manifests as gradual loss of fracture resistance or as an abrupt transition, and whether brittle fracture is activated early or remains inactive until higher hydrogen concentration are reached.

Constraint–hydrogen interaction under current mechanistic understanding. Within the currently accepted set of hydrogen embrittlement mechanisms, crack-tip constraint remains a dominant factor controlling fracture toughness. Higher constraint consistently leads to lower fracture toughness in both air and hydrogen-charged conditions, corresponding to Scenario H1. Hydrogen amplifies the constraint effect by accelerating damage evolution and promoting earlier activation of the brittle fracture mode. The framework does not predict a reversal of the toughness–constraint relationship, indicating that experimentally reported low-constraint/low-toughness behavior (Scenario H3) cannot be rationalized within the current HELP-HESIV-HEDE-based description.

Conceptual limitation and pathway toward a fully unified void-based description. A key limitation of the present H-CGM+ formulation is the heterogeneous treatment of ductile and brittle fracture, with hydrogen-suppressed ductile failure governed by HEVN and hydrogen-assisted brittle failure governed by a cohesive threshold strength. While this approach reflects current state-of-the-art, it is not fully consistent from a micromechanical standpoint. A promising direction is the development of a fully void-based framework in which macroscopically brittle

fracture emerges from extreme localization of hydrogen-induced void nucleation within a thin weakened layer [80]. In such a description, fracture mode becomes a natural outcome controlled by void morphology and void density (or an associated localization length scale), rather than by separate ductile and brittle criteria.

Overall, H-CGM+ provides a robust and extensible platform for mechanistic prediction of hydrogen-assisted fracture toughness and its dependence on crack-tip constraint. The results reinforce the classical role of crack-tip constraint under current mechanistic assumptions, while clearly identifying the limitations of existing models. Progress toward a fully unified void-based description, together with targeted experiments to quantify hydrogen-controlled void nucleation, will be essential for resolving discrepancies with emerging experimental evidence and for establishing reliable fracture toughness assessment methodologies for hydrogen transport and storage infrastructure.

CRediT authorship contribution statement

Jianying He: Writing – review & editing, Supervision, Methodology. **Andrés Díaz:** Writing – review & editing, Methodology. **Haiyang Yu:** Writing – review & editing, Supervision, Methodology. **Karl Etienne Dany Druesnes:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Zhiliang Zhang:** Writing – review & editing, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Vigdis Olden:** Writing – review & editing. **Lise Jemblie:** Writing – review & editing. **Bård Nyhus:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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